1 Introduction and inequalities

Week 1 · Week summary

Core ideas, lesson by lesson

1 Before you start: numbers, sets, and algebra

- $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$, and the irrationals complete $\mathbb{R}$.
- "And" is intersection ($\cap$); "or" is union ($\cup$).
- If $ab = 0$, then $a = 0$ or $b = 0$; never divide by zero.
- $\sqrt{(a^2)} = |a|$, and $|a - b|$ is the distance between a and b.

2 Equations: degree, type, and roots

- Degree = highest power after expanding, moving everything to one side, and simplifying.
- 1st degree $ax + b = 0$ (one root), 2nd $ax^2 + bx + c = 0$ (up to two), 3rd up to three: degree n, at most n roots.
- Zero product rule: $A \cdot B = 0$ means $A = 0$ or $B = 0$. Check each root by substitution.

3 Second-degree equations and the parabola

- $x = (-b \pm \sqrt{\Delta})/2a$ with $\Delta = b^2 - 4ac$: $\Delta > 0$ two roots, $\Delta = 0$ one, $\Delta < 0$ none.
- $a > 0$ opens up (minimum); $a < 0$ opens down (maximum).
- Vertex $V = (-b/2a, -\Delta/4a)$; axis $x = -b/2a$; y-intercept c; x-intercepts are the roots.

4 Radical, absolute value, and fraction equations

- Radicals: isolate, square, then check every answer.
- $|x| = x$ if $x \geq 0$, $-x$ if $x < 0$; $|A| = k \rightarrow A = k$ or $A = -k$.
- Fractions: write the forbidden values first, then multiply by the denominator.

5 From words to equations

- Name the unknown with its unit before writing anything else.
- Translate sentence by sentence; check the answer in the story.
- Discard answers that make no sense: negative lengths, fractions of people.

6 Higher-degree polynomials: Ruffini's rule

- Candidates: ± divisors of the constant (over divisors of the leading coefficient).
- Remainder 0 means a root; the bottom row is the quotient.
- Write 0 for missing powers, and repeat until a quadratic is left.

7 Types of intervals and linear inequalities

- Bounded: (a, b), $[a, b]$, $[a, b)$, $(a, b]$. Infinite: (a, ∞), $[a, \infty)$, $(-\infty, a)$, $(-\infty, a]$.
- [ ] and filled dot = included; ( ) and hollow dot = excluded; ∞ always takes ( ).
- Multiplying or dividing by a negative flips the inequality.

8 Polynomial and rational inequalities: sign tables

- Zero on one side, factor, critical points, one test value per interval.
- $\leq$ or $\geq$ include the numerator's zeros; denominator zeros are always excluded.
- For a quadratic, the sign table is the parabola read along the x-axis.

9 Systems, absolute values by cases, and two variables

- A system keeps the intersection of the solutions.
- Two absolute values: split the line where each inside is zero, solve each case, keep answers inside their case.
- In two variables: $\leq \geq$ solid boundary, $< >$ dashed; test a point to pick the side.

Interval	$x < -4$	$-4 < x < 3$	$x > 3$
Test value	$x = -5$	$x = 0$	$x = 4$
$x + 4$	−	+	+
$x - 3$	−	−	+
$(x + 4)(x - 3)$	+	−	+

> 0



Watch out

Mistake	Fix
Forgetting to flip the sign when dividing by a negative number.	Circle every negative divisor before dividing.
Including a zero of the denominator in the answer.	Mark those points with a hollow dot from the start.
Thinking $-5 > -2$ because $5 > 2$.	Picture the line: -5 is further left, so $-5 < -2$.
Writing $(a + b)^2 = a^2 + b^2$.	The middle term is missing: $(a + b)^2 = a^2 + 2ab + b^2$. Try $a = b = 1$: 4, not 2.
Describing $x^2 > 1$ as "$x > 1$ and $x < -1$".	No number does both; the answer is a union: $(-\infty, -1) \cup (1, \infty)$.
Reading the degree before simplifying.	Expand and cancel first; only then look at the highest power.
Dividing both sides by x and losing the root $x = 0$.	Factor x out instead: $x^2 = 5x \rightarrow x(x - 5) = 0 \rightarrow x = 0$ or $x = 5$.
Simplifying $(4 \pm 2\sqrt{2})/2$ as $4 \pm \sqrt{2}$.	Divide every term: $(4 \pm 2\sqrt{2})/2 = 2 \pm \sqrt{2}$.